Text Classification using Classical ML

Sparse Features vs Contextual Embeddings

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Abstract

This project evaluates classical machine learning techniques for text classification on the AG News dataset. Sparse representations such as Bag-of-Words and TF-IDF are compared against dense contextual embeddings including DistilBERT and GloVe. Multiple classifiers including Logistic Regression, Linear SVM, KNN, and KMeans are benchmarked to study performance, computational trade-offs, clustering quality, and dimensionality reduction behaviour.

1 Introduction

Text classification is one of the most important tasks in Natural Language Processing. The goal of this project is to compare traditional sparse feature extraction techniques with modern contextual embeddings for news-topic classification.

The dataset contains four balanced news categories:

- World
- Sports
- Business
- Sci/Tech

The project investigates:

1. Sparse vs contextual representations
2. Performance of different classifiers
3. Effect of dimensionality reduction
4. Clustering quality using KMeans

2 Dataset and Preprocessing

The experiment uses 8,000 balanced samples from the AG News dataset.

Train-Test Split

- Training Samples: 6,400
- Testing Samples: 1,600

Sparse Feature Preprocessing

- Lowercasing
- Stop-word removal
- Tokenization
- Lemmatization

DistilBERT Preprocessing

DistilBERT embeddings are generated directly from raw text using WordPiece tokenization.

3 Feature Representations

3.1 Sparse Features

- **Bag-of-Words (BoW)**
- **TF-IDF**

3.2 Dense Embeddings

- **DistilBERT-CLS**
- **DistilBERT-Mean**
- **GloVe**

4 Models Used

The following machine learning models are evaluated:

- Logistic Regression
- Linear SVM
- KNN
- KMeans Clustering

5 Results

5.1 Classification Performance

Key Findings

- GloVe + Logistic Regression achieved the highest accuracy.
- Contextual embeddings outperform sparse features overall.
- TF-IDF remains a strong lightweight baseline.
- Dense embeddings significantly improve KNN performance.

6 Dimensionality Reduction

Truncated SVD is applied to sparse TF-IDF features while PCA is applied to dense embeddings.

The experiments show that dimensionality reduction preserves most classification performance while reducing memory and computational cost.

7 Visualizations

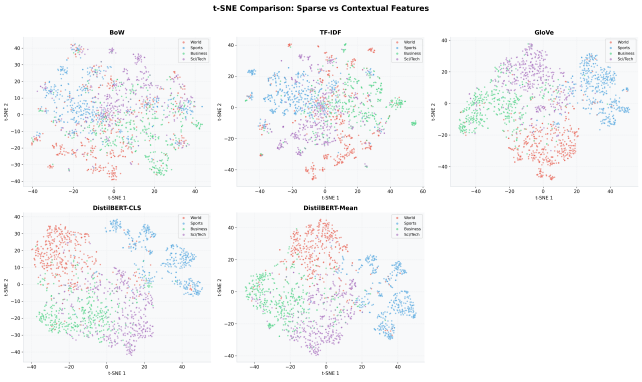


Figure 1: t-SNE projections of different feature representations. Contextual embeddings form tighter and better-separated clusters.

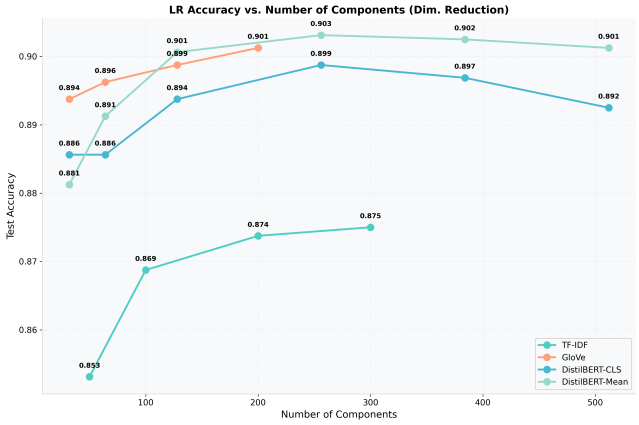


Figure 2: Classification accuracy versus reduced dimensions using PCA/SVD.

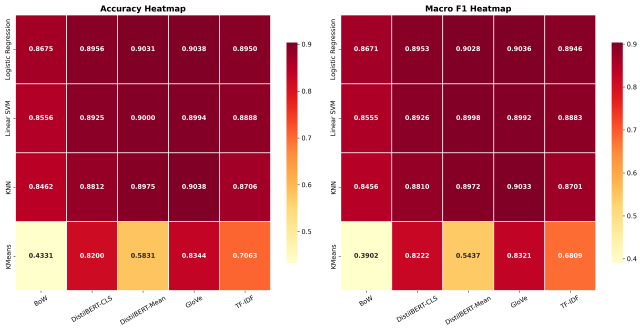


Figure 3: Heatmap showing Accuracy and Macro-F1 across all feature-model combinations.

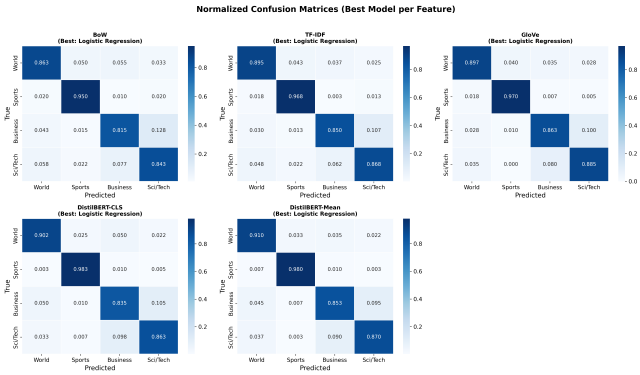


Figure 4: Normalized confusion matrices for the best-performing models.

8 Discussion

The experiments clearly demonstrate the advantage of contextual embeddings over sparse text representations. DistilBERT and GloVe capture semantic relationships more effectively, leading to improved classification accuracy.

However, TF-IDF still provides competitive results with significantly lower computational cost. This makes sparse methods useful for lightweight or resource-constrained applications.

Table 1: Performance comparison across feature representations and classifiers

Feature	Model	Accuracy	Macro F1
BoW	Logistic Regression	0.8675	0.8671
BoW	Linear SVM	0.8556	0.8555
BoW	KNN	0.8462	0.8456
TF-IDF	Logistic Regression	0.8950	0.8946
TF-IDF	Linear SVM	0.8888	0.8883
TF-IDF	KNN	0.8706	0.8701
GloVe	Logistic Regression	**0.9038**	**0.9036**
GloVe	Linear SVM	0.8994	0.8992
DistilBERT-Mean	Logistic Regression	0.9031	0.9028

KNN benefits strongly from dense vector representations because sparse high-dimensional spaces suffer from the curse of dimensionality.

The t-SNE visualizations further confirm that contextual embeddings create more separable feature spaces.

9 Conclusion

1. Contextual embeddings generally outperform sparse representations.
2. TF-IDF remains an efficient and competitive baseline.
3. Dense embeddings significantly improve KNN classification.
4. KMeans captures meaningful topic clusters but performs worse than supervised methods.
5. Dimensionality reduction preserves most classification information while reducing computational cost.

— End of Report —